AI-Powered Tea Quality and Moisture Prediction with Intelligent Auction Price Forecasting

Project ID: 25-26J-292

Project Proposal Report

De Silva P.S – IT22223326

BSc (Hons) Degree in Information Technology (specialized in Information Technology)

Department of Information Technology

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DECLARATION

I declare that this is my own work & this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of my knowledge & belief it does not contain any previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
De Silva P.S	IT22223326	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

_ _

Dr. Nathalie De Silva — Date

Signature of the supervisor

_ _

Ms. Shashini Kumarasinghe — Date

Signature of the co-supervisor

ACKNOLEDGEMENT

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I would also like to thank the members of my group for their cooperation, dedication, and collective effort in making this project.

Finally, I am greatly indebted to my parents for their unconditional love, support, and encouragement, which have been a source of inspiration during this time.

ABSTRACT

Sri Lanka's tea industry is vital to the national economy. Tea exports significantly contribute to foreign revenue and uphold the country's global reputation for quality. However, tea grading and evaluation depend heavily on expert tasters who visually assess samples and give qualitative feedback. While this method works to some degree, it is subjective, slow, and inconsistent. This often results in mis-grading, lower buyer confidence, and inefficiencies in large operations. These evaluations also play an important role in tea auctions, where inconsistencies can directly affect market value and transparency. As global competition and supply chain demands increase, the need for a scalable, data-driven, and standardized grading approach has become more urgent. The study focuses on the subjectivity and inefficiency of current tea grading practices, especially during the auction process where consistency is key. The main goal is to create a web application that uses computer vision to analyze dried tea leaves for quality and produce expert-like comments. The system looks at visual features such as leaf size, twist, and uniformity to classify grades like Orange Pekoe (OP) and Broken Orange Pekoe One (BOP1). The evaluation follows industry standards and auction requirements to ensure clarity and effectiveness. By using computer vision techniques and machine learning algorithms, the application seeks to minimize grading inconsistencies, improve scalability, and deliver reliable feedback similar to what expert tasters provide. The study concludes with a tech-focused method for tea grading that boosts efficiency, promotes fair auction practices, and strengthens Sri Lanka's position in the global tea export market.

Keywords – Tea Grading, Computer Vision, Machine learning

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INTRODUCTION

Background

Tea has been a key agricultural product in Sri Lanka. It brings in foreign currency and helps maintain the island's reputation for high-quality Ceylon tea [1]. Besides direct export income, the industry supports rural jobs and related sectors like packaging and logistics. It also helps define Sri Lanka's brand in the global beverage market [1]. However, essential quality tasks, especially grading, are still mostly done by hand. The evaluation of tea quality is a critical process that directly influences pricing in tea auctions, buyer confidence, and overall brand reputation. Traditionally, tea valuation consists of three major components [2]:

- **Dry Leaf Valuation** – Visual inspection of the processed tea leaves.
- **Liquor Evaluation** – Tasting and visual inspection of brewed tea, assessing color, body, and strength.
- **Infused Leaf Evaluation** – Assessment of the infused or steamed leaves for aroma and uniformity.

These sensory assessments are anchored in standardized brewing protocols for sensory tests (ISO 3103) to ensure reproducibility when preparing liquor for cupping and evaluation [2, 12].

Among these, dry leaf valuation is the first and perhaps the most important stage. It gives a clear sign of overall quality before more testing. This stage mainly looks at the appearance and physical characteristics of the tea leaves. These factors directly affect market grading and price setting at auction [3].

Key descriptors used in dry leaf valuation include:

- **Blackness** – Rich dark color, a sign of quality.
- **Well Twisted / Wiry** – Indicates leaves are carefully rolled and thin in appearance.
- **Bold** – Oversized leaf particles that reduce grade quality.
- **Evenness** – Uniform leaf size, true to grade.
- **Flaky** – Flat or open texture, often undesirable.
- **Make** – Whether the tea is "well-made" and true to grade.
- **Tip** – Fine plucking standard, especially in premium low-grown teas.

Source: Sri Lanka Tea Board's tasting division and related technical pages [5, 13, 15].

Grading into commercial denominations, such as OP, BOP1, and FBOP, depends on leaf style, size, twist, and uniformity. Formal naming is maintained by Sri Lankan industry bodies [3, 11]. At the Colombo Tea Auction, which is one of the world's largest weekly tea auctions, grading consistency affects prices directly. It also influences buyer trust and market transparency [4, 19]. Individual judgment, environmental conditions, and taster fatigue can introduce variability. This can lead to differences in pricing and perceived quality risk [14].

Table 1.1: Key Visual Factors Affecting Tea Grading

Independent Variables	**Influence on Grading**
Leaf Size	High
Color & Texture	High
Leaf Twist	High
Leaf Uniformity	Medium
Fragmentation Level	Medium

Source: Adapted from Sri Lanka Tea Board grade/tasting guidance and tea quality evaluation references [3,5].

This study proposes an AI-powered tea grading system that automates the classification and generation of expert-style comments for dry tea leaves. It focuses on attributes that are important for auction decisions, such as leaf size, twist, appearance, uniformity, color, texture, and fragmentation. By matching the model outputs with Sri Lankan grading terms and cupping practices, the system aims to reduce subjectivity, improve operational efficiency, and boost international buyer confidence. This will help Sri Lanka maintain its competitive edge in premium tea markets [1, 3, 4, 19].

1.1 Literature Survey

1.2.1 Traditional Tea Grading Practices

Tea grading has traditionally depended on the skills of human tasters. They visually evaluate the appearance of dry leaves, the characteristics of the liquor, and the infused leaves to determine quality [1], [2]. While these evaluations are still central to the industry, they are subjective, inconsistent, and require a lot of labor [3], [4]. Changes in environmental conditions, taster fatigue, and personal judgment can greatly affect grading results. This, in turn, impacts auction prices, buyer trust, and brand reputation [5].

Several terms describe dry leaf evaluation, such as leaf blackness, wiriness, evenness, and tip content [2], [6]. However, it has been a constant struggle to maintain consistency in these factors across large operations. As a result, the absence of standardized, reliable grading methods limits the industry's scalability and transparencuy [7].

1.2.2 Early Computational Approaches in Tea Quality Assessment

The first computational efforts in tea quality assessment focused on feature engineering. Researchers extracted hand-crafted descriptors like color histograms, texture features, and shape descriptors from leaf images. They used classifiers like SVM and Random Forest to analyze these features [8], [9]. While these methods showed promise, they had issues with noise, lighting, and leaf differences. This limited their use in real-world grading environments [10].

These problems are similar to earlier work in computer vision, such as facial recognition and object detection. For instance, Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) faced the same challenges [11]. As a result, feature-based grading methods could not adapt well across different tea estates and processing conditions.

1.2.3 Deep Learning and Computer Vision in Tea Grading

The rise of deep learning, especially convolutional neural networks (CNNs), has changed automated tea quality assessment. CNNs can learn visual features directly from raw leaf images. They outperform hand-crafted methods in both accuracy and scalability [6], [12]. Recent studies show that deep learning models can achieve over 95% accuracy in tea grade classification. They can also handle tasks like shoot detection, disease identification, and blending evaluation [7], [13], [14].

For example, Zhao et al. [15] used CNNs for grading fresh tea leaves, showing high reliability in distinguishing plucking standards. Similarly, Guo et al. [16] combined computer vision with blending algorithms to standardize mixture ratios. Wang et al. [6] demonstrated YOLO-based models for real-time grading. These developments highlight AI's ability to replicate expert tasters' judgment while providing consistency, speed, and scalability.

1.2.4 Integration of AI with Sensory Evaluation and Auction Practices

Beyond evaluating dry leaves, there have been efforts to use multiple assessment methods. These methods combine images of dry leaves with the analysis of liquor and infused leaves. Such approaches resemble the workflows of professional tasters more closely and provide complete quality evaluation processes. Studies also suggest that automated systems can improve transparency in auctions and boost buyer confidence. They can reduce variability and ensure traceability in pricing decisions.

However, most previous research focuses on grading from a single perspective, either dry or fresh leaves, with only a little integration of assessments from both dry and infused views. Additionally, the automated generation of comments based on industry grading terms has not been thoroughly explored. This leaves opportunities for innovation in AI-driven evaluation systems that are easy to understand.

1.2.5 Identified Gaps and Research Contribution

While existing studies highlight the potential of AI and computer vision in tea valuation, most focus on liquor quality analysis or multi-stage evaluations of liquor, infused leaves, and dry leaves. However, limited research has specifically targeted dry tea leaf evaluation, especially in assessing important visual features like color, twist, and boldness. Furthermore, there is no established method to verify the accuracy of human tea tasters, which makes maintaining consistency and reliability difficult. The lack of annotated datasets for Sri Lankan tea varieties also hinders the creation of effective automated solutions.

This research aims to fill these gaps by introducing a machine learning framework designed for grading dry tea leaves, with a particular emphasis on Sri Lankan tea. Beyond functioning as a grading tool, the proposed system can assist junior tea tasters as a learning and validation platform, helping to bridge knowledge gaps while ensuring objective and standardized quality assessments.

1.2. Research Gap

Tea grading is a vital process in Sri Lanka's economy. It directly affects auction prices, buyer confidence, and the global reputation of Ceylon tea. Traditionally, expert human tasters have assessed dry tea leaves based on visual traits like color, twist, boldness, uniformity, and fragmentation. Although this method has been in use for decades, it remains subjective. It heavily depends on personal experience and often varies between tasters and sessions.

Recent studies show that most research using Artificial Intelligence (AI) and Machine Learning (ML) for tea quality assessment has concentrated on liquor evaluation, which includes flavor, aroma, and color, or on multi-stage valuation involving brewed and infused leaves. These studies highlight AI's potential to standardize tea evaluation but reveal a significant gap in analyzing dry tea leaves. This step is the first and one of the most crucial parts of grading. Very few efforts have aimed to quantify visual traits like blackness or twist into measurable indices. This leads to a lack of structured and objective frameworks for evaluating dry leaves.

Another important gap is the absence of validation systems for human tea tasters. Right now, there is no objective method to confirm if a senior taster's judgment matches grading standards. This absence of verification lowers transparency and complicates the training of junior tasters. Typically, they learn through apprenticeship without systematic feedback or benchmarks. Existing research seldom explores how automated grading systems could help train tasters, which further widens the gap between traditional methods and modern technology.

Moreover, most datasets available for tea grading are small and often focus on Chinese or Indian tea varieties. This creates an opportunity to develop a dataset that represents Sri Lankan Ceylon teas, which have distinct grading standards and visual traits. The lack of Sri Lanka-specific datasets limits the creation of locally relevant ML and DL models, which in turn affects industrial adoption.

In addition, many previous AI-based grading methods focus solely on classification, assigning grade labels. However, they do not provide interpretable comments that match industry terms like "well twisted," "wiry," or "even." This lack of interpretability reduces trust among industry stakeholders, who want not just automated grading but also explanations using familiar expert language.

In summary, while prior studies have made progress in automating tea quality analysis, there is a clear research gap in:

- Dry leaf–only grading systems, specifically for Sri Lankan tea varieties.
- Validation frameworks for assessing the reliability of human tasters.
- Training support for junior tasters through explainable AI feedback.
- Locally curated datasets that capture the unique characteristics of Ceylon tea.
- Explainable outputs that map ML predictions to Sri Lankan grading vocabulary.

The proposed study seeks to bridge these gaps by developing a **machine vision–based dry tea leaf grading system** that not only automates classification but also generates interpretable expert-style comments. By providing measurable grading criteria, validating human judgments, and supporting training for junior tasters, this system has the potential to enhance transparency, efficiency, and global competitiveness of Sri Lanka's tea industry.

1.5 Research Problem

According to tea industry experts and reports from the Sri Lanka Tea Board, global demand for high-quality Ceylon tea remains significant. Maintaining consistent tea quality is crucial for preserving its international reputation. Traditionally, professional tea tasters evaluate tea quality, particularly through dry leaf assessment. This process relies heavily on human skill, examining factors like color, shape, and level of twist. However, these evaluations are subjective. Results can vary based on the experience, judgment, and even physical condition of individual tasters. Furthermore, there is no established method to confirm the accuracy of these human assessments, leading to uncertainty in quality assurance.

Despite the importance of dry leaf evaluation, most research in tea quality analysis has concentrated on liquor evaluation or chemical composition analysis. The visual assessment of dry leaf quality has received limited technological attention. Current methods lack automation and standardization, which makes it hard to ensure consistency among different evaluators. While expert tasters can generally make reliable judgments, junior tasters struggle to reach the same level of expertise due to the absence of a systematic way to validate their evaluations against objective standards.

Given these issues, the research problem addressed here is:
“How can we develop an automated, ML-based system to objectively evaluate dry tea leaf quality to improve accuracy, reduce subjectivity, and provide a validation framework for both expert and junior tea tasters?”

This research aims to design and implement a system that uses computer vision and machine learning to analyze visual features like color, blackness, and leaf appearance in dry tea leaves. By delivering consistent, explainable, and validated results, the system will support quality assurance in the tea industry and serve as a training tool for junior tea tasters. Unlike traditional subjective methods, this solution will provide an objective, repeatable, and scalable framework that connects human expertise with modern technological validation.

2. OBJECTIVES

2.1 Main Objective

Tea tasting is a skilled process that is vital for assessing the commercial value and overall quality of tea. Traditionally, this process relies on the judgment and knowledge of professional tea tasters, who look at factors like leaf appearance, color, aroma, and liquor quality. However, because it is manual and sensory-based, the process can vary, and it is affected by human bias and difficulties in validation. Junior or less experienced tea tasters often find it hard to match the accuracy of experts. This can lead to unreliable assessments and affect decision-making in the tea industry.

The main goal of this research is to create an AI system for objective evaluation of tea leaf quality, focusing on dry tea leaves. By using machine learning and image analysis, the system seeks to provide a standardized, accurate, and repeatable method for grading tea leaves. This solution not only decreases reliance on subjective human judgment but also offers a tool that can validate and assist professional tasters. It can also serve as a educational resource for junior tasters. In the end, the system promotes greater consistency, fairness, and transparency in evaluating tea quality, helping the tea industry stay competitive worldwide.

2.2 Specific Objectives

- Develop an AI-based tea leaf grading system

Design a system that uses computer vision and machine learning to analyze the visual features of dry tea leaves. This includes leaf size, color, twist, uniformity, and fragmentation to ensure grading meets Sri Lankan tea industry standards and auction needs.

- Provide expert-like feedback and grading comments

Implement a feature that generates interpretative comments in industry-recognized terminology, such as “well twisted,” “bold,” or “even,” to make AI predictions understandable and actionable for both expert and junior tea tasters.

- Support training and validation for junior tea tasters

Create a platform where junior tasters can compare their evaluations with AI-generated results. This will help them learn and validate their judgments against standardized standards.

- Ensure scalable and consistent grading

Develop a solution that can quickly and reliably evaluate large batches of tea leaves. This will reduce subjectivity and variability caused by human factors like fatigue or personal bias.

3. SYSTEM METHODOLOGY

The methodology for this research focuses on creating an AI-powered tea grading system that digitizes the traditionally subjective evaluation process of dry and infused tea leaves. The proposed system, "TeaVision," integrates computer vision, deep learning, and human-expert knowledge to provide consistent, explainable, and scalable grading results. The approach establishes a step-by-step procedure from data acquisition to model deployment, with iterative testing, validation, and expert feedback to ensure alignment with industry standards.

3.1. Application of Key Pillars within the Specialized Knowledge Domain

This study incorporates computer vision, deep learning, and domain knowledge of tea grading:

- **Computer vision** provides the foundation for analyzing visual characteristics of tea leaves. Image processing techniques are applied to extract features like leaf color depth, twist shape, bloom, and uniformity. Controlled lighting and angled macro photography simulate human sensory evaluation conditions.
- **Deep learning** with convolutional neural networks (CNNs) enables automated classification of tea leaf quality, capturing subtle visual cues that human tasters use to assign scores. The system predicts expert-style scores and grading categories such as OP, BOP1, and FBOP.
- **Human-expert knowledge integration** ensures that model outputs correspond to industry-recognized grading standards. This allows the AI system to mimic taster

evaluation and generate interpretable comments for both expert and junior tasters.

3.2. Application of Technologies in the Field of Tea Grading

The proposed module leverages current frameworks and hardware to ensure precision and efficiency:

- **Image acquisition setup**: A light-controlled imaging box with a 45-degree camera angle captures dry and wet tea leaf samples under standardized conditions. Both macro and angled imaging simulate visual inspection by human tasters.
- **Feature extraction**: Image processing techniques extract visual quality indicators such as leaf color, twist, bloom, fragmentation, and uniformity. Dual-view assessment (dry and infused leaf images) is incorporated to enhance grading accuracy.
- **Deep learning classification**: CNN-based models are trained to classify tea leaf quality and generate taster-style scores on a 1–10 scale, reflecting professional sensory evaluation practices.
- **User interface**: A simple GUI visualizes sample evaluation results in real-time, allowing easy comparison between AI predictions and human taster feedback.

3.3. How to Carry Out the Project

The project will adopt an iterative development lifecycle:

- **Requirement analysis and design** – Define technical specifications for imaging setup, feature extraction, CNN architecture, and user interface.
- **Image dataset collection** – Capture high-quality images of diverse tea leaf samples (dry and wet) with known expert taster scores. Ensure consistency with controlled lighting and standardized angles.
- **Feature extraction and preprocessing** – Process images to extract key visual indicators, normalize data, and augment datasets for robust model training.
- **Model training and optimization** – Train CNN models to classify grades and generate expert-style scores, iteratively optimizing for accuracy and generalization.
- **Validation and expert feedback** – Compare model predictions with human taster evaluations, refine model performance, and ensure consistency with grading standards.
- **System integration and GUI development** – Develop a front-end tool to visualize results, display grading comments, and provide a scoring dashboard.

- **Evaluation** – Assess model performance quantitatively (accuracy, consistency) and qualitatively through expert validation, ensuring scalability and reliability for auction-ready grading.

3.4. Tasks and Sub-Tasks Identified to Meet Objectives

1. **Imaging setup development**
    - Design light-controlled box and camera positioning for dry and wet leaf capture.
    - Standardize imaging protocols for lighting, angle, and distance.
2. **Dataset collection and annotation**
    - Capture diverse tea samples with known taster scores.
    - Annotate images with grading categories and taster-style scores.
3. **Feature extraction and preprocessing**
    - Extract visual features: color depth, twist, bloom, uniformity, and fragmentation.
    - Apply preprocessing: normalization, augmentation, and dual-view alignment.
4. **Model development**
    - Design CNN architecture for grade classification.
    - Train model to generate scoring aligned with expert evaluation.
    - Optimize inference for real-time assessment.
5. **System integration and GUI development**
    - Build interactive interface to visualize AI predictions and comments.
    - Enable comparison with human evaluations for training and validation.
6. **Testing and evaluation**
    - Validate model accuracy with expert feedback.
    - Evaluate consistency, scalability, and usability for real-world auction scenarios.

3.5. Data Requirements and Collection Methods

The module requires a comprehensive image dataset representing Sri Lankan tea varieties:

- **Dry and wet leaf images** captured under standardized lighting and angles.
- **Expert-annotated grading scores** for each sample.
- **Diversity in tea types and grades** to ensure generalizability.
- **Preprocessing** includes leaf segmentation, color correction, alignment, and augmentation to enhance robustness.

Data sources may include publicly available tea leaf image datasets as well as controlled data collected from pilot studies at tea estates. Captured images will be processed using image segmentation, alignment, and augmentation techniques such as rotation, cropping, and brightness adjustment to increase model robustness. All test data will be anonymized, storing only extracted features rather than raw images, ensuring privacy and compliance with data handling best practices.

3.6. Project Requirements

3.6.1. Functional Requirements

• Accurately grade dry and wet tea leaves

The system must use computer vision and deep learning algorithms to assess tea leaf quality in real time from images captured under controlled lighting and angles. The grading model should be trained on a large, annotated dataset to ensure robustness against variations in leaf type, size, color, twist, and fragmentation. Accurate grading is essential for maintaining auction standards, ensuring fairness, and generating reliable taster-style scores.

• Generate interpretable expert-style comments

The system should produce human-readable comments corresponding to visual quality indicators, such as “well twisted,” “bold,” or “even.” This functionality ensures that both expert and junior tasters can understand, validate, and learn from the AI predictions.

• Integrate seamlessly with scoring and reporting tools

The component must provide outputs in a structured format suitable for visualization, reporting, or further analysis. Integration with GUI or dashboard tools should be smooth, enabling real-time display of grading results, scores, and interpretative comments.

• Support batch processing and real-time evaluation

The system should be capable of evaluating single samples in real time as well as processing large batches of tea leaves for auction-ready grading. This dual-mode capability ensures operational efficiency and reliability for both small-scale testing and large-scale industrial applications.

• Automatically process multiple samples simultaneously

The system must handle multiple tea leaf images at once, providing consistent grading and comments for each sample. This is essential to maintain speed and accuracy in high-throughput environments such as tea estates or auction settings.

3.7. User and System Requirements

3.7.1. User Requirements

• Provide a simple-to-use interface for tea graders and estate staff

The system must have an intuitive interface that allows tea tasters, quality controllers, and estate managers to assess dry tea leaf quality without requiring technical expertise. It should provide clear visualizations of grading results, including scores, expert-style comments, and leaf feature highlights in real time.

• Give rapid results with minimal setup

Users should be able to capture and evaluate dry leaf samples quickly using a controlled imaging setup. The system must operate with standard cameras and general-purpose computing hardware, providing timely grading results to support auction preparation and quality monitoring.

• Ensure transparency and trust

The system should allow users to understand how grading results are derived, showing feature analysis (e.g., leaf twist, blackness, uniformity) alongside predicted grades. This ensures graders trust the automated scoring while maintaining alignment with industry standards.

3.7.2. System Requirements

• High computational efficiency with optimized inference speed

The grading system must analyze dry leaf images quickly and accurately. Models should be optimized using techniques such as model pruning or quantization to enable real-time scoring on mid-range computing hardware.

• Compatibility with standard image input formats

The system should accept images captured via common cameras, DSLR, or smartphone cameras. Inputs must support various resolutions and lighting conditions without requiring specialized imaging equipment.

• Scalable and deployable architecture

The solution must be scalable to process multiple samples simultaneously, supporting both small-scale estate operations and large-scale auction preparation. It should allow integration with dashboards or reporting systems for centralized monitoring.

3.7.3. Non-Functional Requirements

• **Accuracy and robustness**

The model must deliver consistent grading performance across diverse dry leaf samples, minimizing misclassifications. It should maintain reliability under variations in lighting, leaf orientation, and fragmentation.

• **Data privacy**

The system must not store raw leaf images unnecessarily. Only processed features and anonymized data should be retained for training, evaluation, or reporting, ensuring compliance with ethical and data-handling standards.

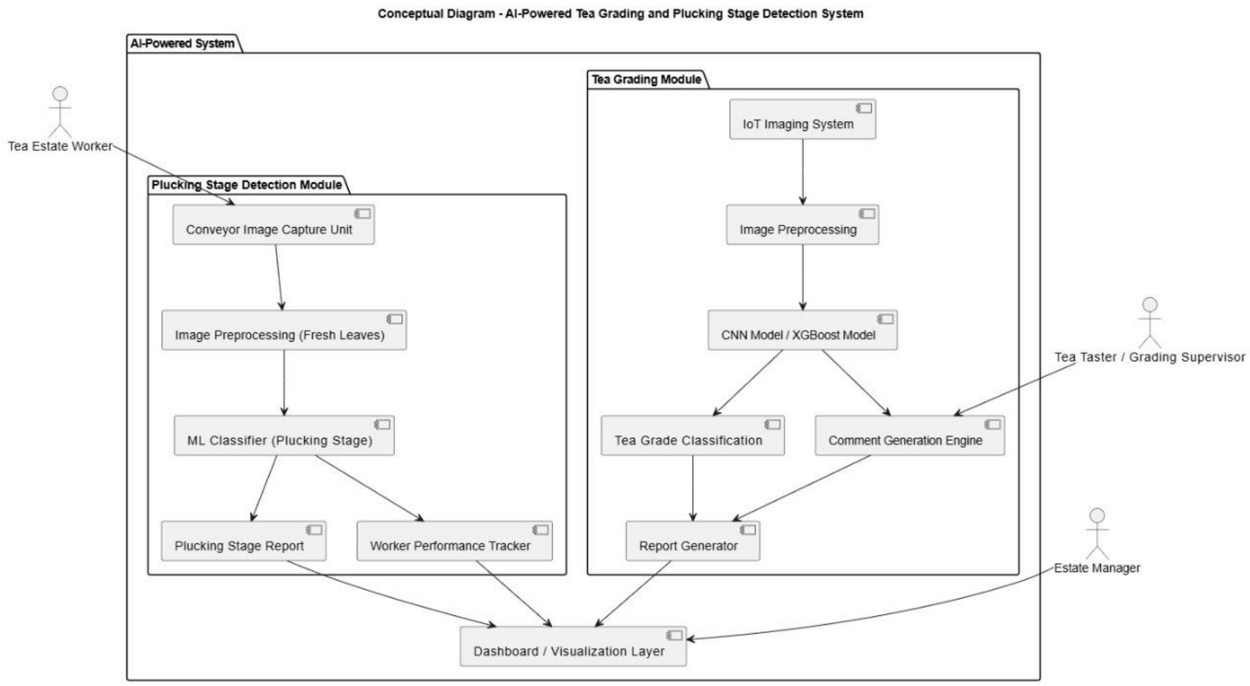
• **Reliability**

The grading tool should operate continuously without frequent interruptions. Error logging, automatic recovery, and failover mechanisms must be implemented to ensure uninterrupted service during high-volume operations.

• **Flexibility**

The system architecture should support easy updates, integration of improved models, or additional visual feature extraction capabilities. This ensures the tool evolves with future advancements in AI-driven tea grading without requiring complete redevelopment.

1.1.1 Use Case Scenarios (tentative)



1.1.2 Test Cases (tentative)

Table 3.7: Attempt Writing Assessement Test Case

<table>
<tr><td colspan="3">Test case ID: TLG_Test_01</td><td colspan="3">Test designed by: De Silva P.S</td></tr>
<tr><td colspan="3">Test title: Perform dry tea leaf grading</td><td colspan="3">Test designed day: 2025/8/30</td></tr>
<tr><td colspan="3">Test priority (High/Medium/Low): High</td><td colspan="3">Test executed by: De Silva P.S</td></tr>
<tr><td colspan="3">Module name: Grading & Feedback Screen</td><td colspan="3">Test executed day: 2025/10/10</td></tr>
<tr><td colspan="6">Description: Attempt grading of a dry tea leaf sample using the system and receive quality score with expert-style feedback.</td></tr>
<tr><td colspan="6">Pre-conditions: Ensure image capture device or uploaded dataset is accessible</td></tr>
<tr><td colspan="6">Testing steps:
1. Login to the system as an authorized user.
2. Navigate to the main menu.
3. Select Dry Leaf Grading option.
4. Upload/capture dry tea leaf image.
5. Wait for grading to complete.
6. Click “View Result” button to review quality score and expert-style comments.</td></tr>
<tr><td>Test ID</td><td>Test Inputs</td><td>Expected Output</td><td>Actual Output</td><td>Result (Pass/Fail)</td><td>Comments</td></tr>
<tr><td>TLG _Test_01</td><td>Upload/capture dry tea leaf image</td><td>Receive grading results with detailed quality indicators (e.g., twist, size, blackness, bloom) and expert-style comments.</td><td>Receive grading results with detailed quality indicators and expert-style comments.</td><td>Pass</td><td>Grading completed successfully. Accurate prediction generated by ML model..</td></tr>
</table>

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